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Fluid pumps

Abstract

In accordance with at least one aspect of this disclosure, embodiments of fluid pumps, pump cases, valve bodies, and volutes are disclosed herein. Embodiments of methods for manufacturing fluid pumps, pump cases, valve bodies, and volutes are also disclosed herein. Embodiments can include additive manufacturing, for example. Certain embodiments can include additively manufacturing a pump case in a tilted orientation, utilizing only coincidental support structure having a unique shape, and with teardrop shaped volute and/or valve body.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/359,441, filed on Jun. 25, 2021, the entire content of which is herein incorporated by reference.

FIELD

(1) This disclosure relates to fluid pumps (e.g., augmenter fuel pumps for aircraft).

BACKGROUND

(2) Existing fuel pumps for aircraft are molded and machined. Traditional fuel pump designs and methods make it difficult, if not impossible, to additively manufacture certain fuel pumps to be useful and economical over traditional manufacturing methods, for example.

(3) Conventional methods and designs have generally been considered satisfactory for their intended purpose. However, there is still a need in the art for improved fluid pump designs and methods of manufacturing. The present disclosure provides a solution for this need.

SUMMARY

(4) In accordance with at least one aspect of this disclosure, a method for manufacturing a fluid pump can include additively manufacturing a fluid pump case in a tilted orientation without any non-build plate support structure in any internal apertures of the fluid pump case except for coincidental support structure only in one or more coincidental locations that are already intended to be subjected to post process milling unrelated to the coincidental support structure. In certain embodiments, the method can include milling the one or more coincidental locations to a final shape and/or finish, and in the process, eliminating the coincidental support structure.

(5) The one or more coincidental locations can be one or more valve bodies of the fluid pump case configured to house valve components. The tilted orientation can include a tilt in at least two axes.

(6) A Cartesian build plate coordinate system can be defined relative to a build plate. The Cartesian build plate coordinate system can include a build plate G-axis, a build plate F-axis, and a build plate H-axis. The build plate can have a planar shape define in a build plate G-F plane. A Cartesian part coordinate system can be defined relative to the fluid pump case, the Cartesian part coordinate system can include a part G-axis, a part F-axis, and a part H-axis.

(7) In certain embodiments, the tilted orientation can be such that the part H-axis is tilted relative to the build plate H-axis by about 40 degrees to about 45 degrees in the build plate F-H plane, and the part G-axis is tilted from the build plate G-axis by about 50 degrees to about 55 degrees in the build plate G-F plane.

(8) The tilted orientation can be such that the part H-axis is tilted relative to the build plate H-axis by about 43.6 degrees in the build plate F-H plane, and the part G-axis is tilted from the build plate G-axis by about 52.4 degrees in the build plate G-F plane. The fluid pump case can include a plurality of openings defining a plurality of parallel axes that are parallel with the part H-axis.

(9) The fluid pump case can include at least one opening defining a non-parallel axis that is not parallel with the part H-axis. In certain embodiments, the at least one non-parallel axis can be orthogonal to the parallel axes. Coincidental support structure can be built within at least one of the plurality of openings.

(10) In accordance with at least one aspect of this disclosure, a non-transitory computer readable medium, comprising computer executable instructions configured to cause a computer to execute a method, the method including operating an additively manufacturing machine to build a fluid pump case layer by layer in a tilted orientation such that no non-build plate support structure is used in any internal apertures of the fluid pump case except for only coincidental support structure in one

or more coincidental locations that are already intended to be subjected to post process milling unrelated to the coincidental support structure. The method can otherwise be the same or similar to any method disclosed herein, e.g., as described above. Any suitable other method(s) and/or portion(s) thereof are contemplated herein.

(11) In accordance with at least one aspect of this disclosure, an additive manufacturing system can have a non-transitory computer readable medium as disclosed herein, e.g., as described above. In accordance with at least one aspect of this disclosure, embodiments can include a means for additively manufacturing a fluid pump case.

(12) In accordance with at least one aspect of this disclosure, a fuel pump for an aircraft can include a fluid pump case formed by any suitable method disclosed herein, e.g., as described above. The fluid pump can include one or more fluid pump components (e.g., impeller, one or more valves, etc.) disposed within the fluid pump case.

(13) In accordance with at least one aspect of this disclosure, an additively manufactured valve body for a fluid pump case can include an interior surface, a plurality of interior annular features integrally formed with and extending radially inwardly from the interior surface and positioned apart in an axial direction relative to each other, and a support structure formed integrally with and connecting a lower-in-build annular feature to a higher-in-build annular feature. In certain embodiments, the support structure can include a filled-in Y-shape.

(14) For example, the support structure can include a stem and a wide portion extending from the stem. The stem of the support structure is integrally formed with and extends from a lower-in-build portion of the lower-in-build annular feature. The wide portion can extend from the stem and can be integrally formed with and connected to a higher-in-build portion of the higher-in-build annular feature.

(15) The valve body can include a support structure for each annular feature. Each support structure can extend between each successive pair of annular features. In certain embodiments, the valve body can be constructed at an angle between the axial direction and a build plate surface of greater than about 45 degrees.

(16) In accordance with at least one aspect of this disclosure, an additively manufactured fluid pump case can include a valve body. For example, the valve body can be as disclosed herein, e.g., as described above.

(17) In accordance with at least one aspect of this disclosure, a method can include additively manufacturing a valve body of fluid pump case as disclosed herein, e.g., as described above. The method can also include milling away the support structure while finish-milling the valve body.

(18) In certain embodiments, additively manufacturing can be or include laser power bed fusion. The method can include removing powder from within the valve body before milling through openings defined by the support structure. The method can include inserting a valve into the valve body. The method can include any other suitable method(s) and/or portions thereof. Embodiments can include a means for additively manufacturing a valve body as disclosed herein, e.g., as described above.

(19) In accordance with at least one aspect of this disclosure, an additively manufactured valve body for a fluid pump case can include an interior surface defining a main channel along an axis in an axial direction. The channel can have a non-uniform cross-section configured to allow the valve body to be additively manufactured without support structure.

(20) In certain embodiments, the non-uniform cross-section can have a teardrop shape. For example, the interior surface can include a curved lower-in-build portion and a ramped higher-in-build portion that forms a V-shape extending from the curved lower-in-build portion. The curved lower-in-build portion can have a semicircle cross-section. The ramped higher-in-build portion can include a curved peak. In certain embodiments, the valve body can be additively manufactured such that the axis forms an angle of about 45 degrees or less with a build plate surface on which the valve body is built.

- (21) The interior surface can further define one or more annular grooves extending radially outwardly from the main channel. The one or more annular grooves include an asymmetric groove profile.
- (22) A higher-in-build portion of the one or more annular grooves can include a curved shape in the axial direction, and a lower-in-build portion of the one or more annular grooves can include a straight shape in the axial direction.
- (23) In certain embodiments, the one or more annular grooves can be two annular grooves positioned apart in an axial direction relative to each other. Any suitable number of annular grooves are contemplated herein.
- (24) In accordance with at least one aspect of this disclosure, an additively manufactured fluid pump case can include a valve body as disclosed herein, e.g., as described above. In accordance with at least one aspect of this disclosure, a method of manufacturing a fluid pump case can include additively manufacturing a valve body as disclosed herein, e.g., as described above, and machining the main channel to have a different cross-section to allow the valve body to receive one or more valve components while leaving the one or more annular grooves.
- (25) In accordance with at least one aspect of this disclosure, embodiments can include fluid pump case means for use as a fluid pump housing, the fluid pump case means having valve body means configured to allow additive manufacturing of the fluid pump case without building support structure in the valve body.
- (26) In accordance with at least one aspect of this disclosure, a fluid pump case can include an interior volute surface defining a volute channel. The interior volute surface can include a first portion having a D-shaped cross-section, a second portion having a teardrop shaped cross-section downstream of the first portion, and a transition portion between the first portion and the second portion having a changing cross-section.
- (27) The teardrop cross-section can include a curved lower-in-build portion and a ramped higher-in-build portion that forms a V-shape extending from the curved lower-in-build portion. The curved lower-in-build portion can have a semicircle cross-section. The ramped higher-in-build portion can have a curved peak.
- (28) The second portion can be shaped such that it is additively manufactured without support structure. For example, a centerline of the peak along the length of the second portion can be coplanar with a vertical build direction in build such that the teardrop shape is symmetric about a vertical plane.
- (29) The first portion can have an expanding flow area in a flow direction toward the second portion. The second portion can have an expanding flow area or constant flow area in the flow direction.
- (30) The volute channel can further include a reducing portion downstream of the second portion. The reducing portion can include a teardrop shape cross-section. The reducing portion can be connected to a valve channel defined by a valve body.
- (31) In accordance with at least one aspect of this disclosure, a method can include additively manufacturing a fluid pump case to include an interior volute surface defining a volute channel as disclosed herein, e.g., as described above. Additively manufacturing can include aligning a centerline of the peak along the length of the second portion to be coplanar with a vertical build direction in build such that the teardrop shape is symmetric about a vertical plane. In accordance with at least one aspect of this disclosure, embodiments can include volute means for a fluid pump case.
- (32) These and other features of the embodiments of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description taken in conjunction with the drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1A is a perspective view of an embodiment of a fluid pump case in accordance with this disclosure;

(3) FIG. 1B is an underside perspective view of the embodiment of FIG. 1A;

(4) FIG. 1C is a perspective view of the embodiment of FIG. 1A shown relative to a build plate in an embodiment of a tilted build orientation for additive manufacturing, illustrating a location of a build plate coordinate system and a part coordinate system;

(5) FIG. 1D is a front elevation view of the embodiment of FIG. 1C, showing an embodiment of an angle defined between a first pair of axes of the coordinate systems;

(6) FIG. 1E is a top down plan view of the embodiment of FIG. 1D, showing an embodiment of an angle between another pair of axes of the coordinate systems;

(7) FIG. 1F is another perspective view of the embodiment of FIG. 1C;

(8) FIG. 1G is another perspective view of the embodiment of FIG. 1C;

(9) FIG. 1H is another perspective view of the embodiment of FIG. 1C;

(10) FIG. 1I is a cross-sectional perspective view of the embodiment of FIG. 1C;

(11) FIG. 1J is a cross-sectional perspective view of a portion of the embodiment of FIG. 1C;

(12) FIG. 1K is a cross-sectional view of the embodiment of FIG. 1C, shown sectioned in a plane approximately parallel to that of FIG. 1I, looking in an opposite direction;

(13) FIG. 1L is a cross-sectional view of a portion of the embodiment of FIG. 1C, sectioned in a plane approximately orthogonal to that of FIG. 1K.

(14) FIG. 1M is a top down plan view of the embodiment of FIG. 1A;

(15) FIG. 1N is an elevation view of the embodiment of FIG. 1A;

(16) FIG. 1O is a bottom up plan view of the embodiment of FIG. 1A;

(17) FIG. 1P illustrates views of an embodiment of a fluid pump assembled utilizing an embodiment of a pump case of FIG. 1A, and indicating locations of certain components, inlets, and outlets;

(18) FIG. 2A is an elevation view of an embodiment of a valve body (e.g., for a cooling flow return valve) of a fluid pump case of FIG. 1A in accordance with this disclosure, e.g., as shown in section in FIG. 1I;

(19) FIG. 2B is a cross-sectional view of the embodiment of FIG. 2A;

(20) FIG. 2C is a schematic view of the embodiment shown in FIG. 2B, illustrating a powder flow path through openings defined by the coincidental support structure;

(21) FIG. 2D is a cross-sectional perspective view of the embodiment of FIG. 2A, showing an embodiment of a build angle between a center axis of the valve body and a surface of a build plate;

(22) FIG. 2E illustrates an embodiment of a valve body after machining in accordance with this disclosure, showing the coincidental support structure removed;

(23) FIG. 3A is a cross-sectional perspective view of an embodiment of another valve body (e.g., for a discharge valve) of the fluid pump case of FIG. 1A in accordance with this disclosure, e.g., as shown in section in FIG. 1L;

(24) FIG. 3B is a cross-sectional view of the embodiment of FIG. 3A, showing an embodiment of a build angle between a center axis of the valve body and a surface of a build plate;

(25) FIG. 3C is a cross-sectional view of the embodiment of FIG. 3A, illustrating a powder flow path through the valve body built without support structure as shown;

- (26) FIG. 3D an embodiment of a valve body after machining in accordance with this disclosure, showing internal shape changed compared to the shape as additively manufactured;
- (27) FIG. 4A is a perspective view of an embodiment of a volute, e.g., of the fluid pump case of FIG. 1A, shown in negative and in isolation;
- (28) FIG. 4B shows the embodiment of FIG. 4A within a fluid pump case for context;
- (29) FIG. 4C is a cross-sectional view of a volute channel defining the volute of FIG. 4A, showing a teardrop shaped portion thereof;
- (30) FIG. 4D is a cross-sectional view of the volute channel of FIG. 4C, shown sectioned in a plane orthogonal to the of FIG. 4C;
- (31) FIG. 4E is a cross-sectional view of the volute channel of FIG. 4C, showing a reducing portion of the volute channel connecting to a valve channel of a valve body;
- (32) FIG. 4F is a cross-sectional view of the volute channel of FIG. 4C, shown oriented in a build orientation; and
- (33) FIG. 4G is a cross-sectional view of the volute channel of FIG. 4C shown oriented in a build orientation.

DETAILED DESCRIPTION

(34) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, an illustrative view of an embodiment of a fluid pump case in accordance with the disclosure is shown in FIGS. 1A and 1s designated generally by reference character **100**. Other embodiments and/or aspects of this disclosure are shown in FIGS. 1B-4F.

(35) FIGS. 1A-1P show various views of an embodiment of a fluid pump case **100** to illustrate the 3-dimensional structure of the fluid pump case **100** and an embodiment of a build orientation thereof. FIG. 1A is a top perspective view of an embodiment of a fluid pump case **100** in accordance with this disclosure. FIG. 1B is an underside perspective view of the embodiment of FIG. 1A. FIG. 1C is a perspective view of the embodiment of FIG. 1A shown relative to a build plate in an embodiment of a tilted build orientation for additive manufacturing, also illustrating a location of a build plate coordinate system and a part coordinate system. FIG. 1D is a front elevation view of the embodiment of FIG. 1C, showing an embodiment of an angle defined between a first pair of axes of the coordinate systems. FIG. 1E is a top down plan view of the embodiment of FIG. 1D, showing an embodiment of an angle between another pair of axes of the coordinate systems. FIG. 1F is another perspective view of the embodiment of FIG. 1C. FIG. 1G is another front top perspective view of the embodiment of FIG. 1C. FIG. 1H is a rear perspective view of the embodiment of FIG. 1C. FIG. 1I is a cross-sectional perspective view of the embodiment of FIG. 1C, taken at a diagonal line relative to the build plate. FIG. 1J is a cross-sectional perspective view of a portion of the embodiment of FIG. 1C. FIG. 1K is a cross-sectional view of the embodiment of FIG. 1C, shown sectioned in a plane approximately parallel to that of FIG. 1I, looking in an opposite direction. FIG. 1L is a cross-sectional view of a portion of the embodiment of FIG. 1C, sectioned in a plane approximately orthogonal to that of FIG. 1K. FIG. 1M is a top down plan view of the embodiment of FIG. 1A. FIG. 1N is an elevation view of the embodiment of FIG. 1A. FIG. 1O is a bottom up plan view of the embodiment of FIG. 1A. FIG. 1P illustrates views of an embodiment of a fluid pump assembled utilizing an embodiment of a pump case of FIG. 1A, and indicating locations of certain components, inlets, and outlets.

(36) In accordance with at least one aspect of this disclosure, referring generally to FIGS. 1A-1P, a method for manufacturing a fluid pump (e.g., as shown assembled in FIG. 1P) can include additively manufacturing a fluid pump case **100** (e.g., as shown in FIGS. 1A and 1B) in a tilted orientation (e.g., as shown in FIGS. 1C, 1D, 1E, 1F, 1G, 1H, 1I, 1J, 1K, and 1L) such that no non-build plate support structure (i.e., build plate support structure being support structure that is attached to the build plate) is used in any internal apertures of the fluid pump case **100** except for only coincidental support structure (e.g., support structure **103a**, **103b**, **103c** as shown in FIGS. 1I

and **1M**) in one or more coincidental locations **101a, b, c** (e.g., as shown in FIGS. **1A** and **1M**) that are already intended to be subjected to post process milling unrelated to the coincidental support structure. For example, such locations can be locations where post process milling (e.g., any suitable subtractive process) would be required (e.g., for a valve body **200**) to achieve certain surface finishes and/or tighter tolerances than available with additive manufacturing (e.g., laser powder bed fusion). Various views of an embodiment of the tilted orientation of a suitable embodiment of the pump case **100** is shown in FIGS. **1C-1L**.

(37) Certain embodiments may include build plate support structure where desired (e.g., for facilitating the tilted orientation). However, certain embodiments may not require certain support structure (e.g., any, build plate support structure, or other support structure) and support structure may not be desired, in which case build plate support structure and/or other support structure can be avoided.

(38) In certain embodiments, the method can include milling the one or more coincidental locations **101a, b, c** to a final shape and/or finish, and in the process, eliminating the coincidental support structure. Any other suitable post processing in addition to or alternative to milling are contemplated herein.

(39) The one or more coincidental locations **101a, b, c** can be one or more valve bodies of the fluid pump case **100** configured to house valve components. The valve bodies can be portions of a fluid pump case **100** (e.g., integral in the pump case **100** as shown) that are configured to house valve components to form a valve, for example.

(40) As illustrated in FIGS. **1C-1E**, the tilted orientation can include a tilt in at least two axes. A Cartesian build plate coordinate system **105** can be defined relative to a build plate **107**. The Cartesian build plate coordinate system **105** can include a build plate G-axis, a build plate F-axis, and a build plate H-axis, e.g., as shown. The G, F, and H axes are orthogonal as shown. The H-axis can be a vertical axis, for example. As shown, the build plate **107** can have a planar shape defined in a build plate G-F plane (the plane defined between the build plate G and F axes).

(41) A Cartesian part coordinate system **109** can be defined relative to the fluid pump case **100**, e.g., as shown. Similar to the build plate coordinate system **105**, the Cartesian part coordinate system **109** can include a part G-axis, a part F-axis, and a part H-axis (e.g., each axis being orthogonal with the others, the H-axis being the vertical axis). As shown, the part H-axis can be aligned with a center axis of a first opening **111a** (e.g., a pump inlet).

(42) In certain embodiments, as shown in FIG. **1D**, the tilted orientation can be such that the part H-axis is tilted relative to the build plate H-axis by about 40 degrees to about 45 degrees in the build plate F-H plane. As shown in FIG. **1E**, the part G-axis can also be tilted from the build plate G-axis by about 50 degrees to about 55 degrees in the build plate G-F plane. In the embodiment shown, the tilted orientation can be such that the part H-axis is tilted relative to the build plate H-axis by about 43.6 degrees in the build plate F-H plane, and the part G-axis is tilted from the build plate G-axis by about 52.4 degrees in the build plate G-F plane.

(43) The fluid pump case **100** can include a plurality of openings **111b, 111c, 111d, 111e** defining a plurality of parallel axes (e.g., center axes) that are parallel with the part H-axis. The fluid pump case **100** can include at least one opening **113** defining a non-parallel axis (e.g., the center axis thereof) that is not parallel with the part H-axis. In certain embodiments, the at least one non-parallel axis of the at least one opening **113** can be orthogonal to the parallel axes of openings **111a, b, c, d, e**. Coincidental support structure can be built within at least one of the plurality of openings, e.g., openings **111b, 111d, 111e**.

(44) In accordance with at least one aspect of this disclosure, a non-transitory computer readable medium can include computer executable instructions configured to cause a computer to execute a method. The method can include operating an additively manufacturing machine to build a fluid pump case layer by layer in a tilted orientation such that no non-build plate support structure is used in any internal apertures of the fluid pump case except for only coincidental support structure

in one or more coincidental locations that are already intended to be subjected to post process milling unrelated to the coincidental support structure. The method can be the same or similar to any method disclosed herein, e.g., as described above. Any suitable other method(s) and/or portion(s) thereof are contemplated herein.

(45) In accordance with at least one aspect of this disclosure, an additive manufacturing system can have a non-transitory computer readable medium as disclosed herein, e.g., as described above. In accordance with at least one aspect of this disclosure, embodiments can include a means for additively manufacturing a fluid pump case. Embodiments can include fluid pump case means, e.g., as shown in the figures and/or described above.

(46) In accordance with at least one aspect of this disclosure, a fuel pump **99** (e.g., as shown in FIG. **1P**) for an aircraft can include a fluid pump case, e.g., fluid pump case **100** formed by any suitable method disclosed herein, e.g., as described above. The fluid pump **99** can include one or more fluid pump components known to those having ordinary skill in the art disposed within the fluid pump case **100**, e.g., an ejector inlet **150**, a shut off solenoid **152**, a BSD outlet **154**, a cooling flow return **156**, an augmentor discharge **158**, a main flow inlet **160**, an ejector outlet **164**, a selector valve **166**, a pressure temperature sensor **168**, an impeller (not shown), and any other suitable pump components appreciated by those having ordinary skill in the art (e.g., for an augmentor fuel pump for aircraft).

(47) To aid in defining the shown embodiment of a build orientation, below provided in Table 1 is an embodiment of positions of noted points shown in FIG. **1C** relative to the build plate coordinate system. Any other suitable positions that achieve similar results are contemplated herein. The positions are shown in inches.

(48) TABLE-US-00001 TABLE 1 CO-ORDINATES POINTS F-BASIC G-BASIC H-BASIC A
-6.0150 -2.9647 11.6542 B -8.1721 -1.3031 10.1004 C -2.4793 .5027 4.1281 D .0000 .0000
.0000

(49) An embodiment of diameters of openings in the fluid pump case **100** is shown below in Table 2. Any other suitable dimensions are contemplated herein. The dimensions below are reflected in inches.

(50) TABLE-US-00002 TABLE 2 Points Diameter 1 Ø 1.678 ± .005 2 Ø 1.730 3 Ø .435 4 Ø 1.310
5 Ø 1.350 6 Ø .500 7 Ø .311 8 NOT APPLICABLE

(51) Embodiments can include a certain shape (e.g., of one or more channels) and a tilted orientation thereof to allow building with additive manufacturing efficiently. Embodiments can include an orientation of build to avoid extra support structure that would not be machined away with already existing machining. For example, each valve body in a traditional fluid pump case can be milled after traditional casting to provide a final shape and a final finish. Such post processing can be required for additive manufacturing as well. Thus, each valve body can have support structure added therein without adding additional steps of milling. Accordingly, the tilted orientation can be selected such that support structure internal to the device is only utilized where milling is necessary.

(52) Embodiment of additive manufacturing include laser powder bed fusion. Any other suitable process is contemplated herein. In certain embodiments, supports are not permissible on any internal surfaces of the part that are not already going to be machined away. Certain applications (e.g., for aircraft fuel pumps) may require the pump case to be clean, burr free, and powder free, and some surfaces may require a machined finish while others may not.

(53) In certain embodiments, the recoater direction (e.g., the direction a powder recoater of a powder bed fusion machine moves to recoat) can be in the positive F-axis direction of build plate coordinate system (e.g., the direction arrow is pointing in the FIG. **1C**). The build direction can be the positive H-axis direction of build plate coordinate system (e.g., vertical).

(54) Embodiments can allow additive manufacturing of an augmentor fuel pump case which was initially designed and optimized for a casting manufacturing process. Castings generally have high

startup cost and long lead times. Additive manufacturing (AM) can be an enabling technology for this part to save costs in the supply chain and maintain or improve part performance when properly redesigned for both performance, cost, and manufacturability.

(55) The AM build configuration as disclosed herein can be an intermediate step, as there can be post AM processes that are used to create the final part (e.g., machining the valve bodies to finish while removing any support structure). The orientation, support structure, and modifications of embodiments of an additively manufactured fluid pump case are unique compared to a traditional cast pump case.

(56) Through utilizing a tilted orientation as disclosed above, and design features that avoid the use of internal support structures in hard to reach or traditionally unfinished locations, embodiments of a method can enable improved manufacturing of an augmentor fuel pump through additive manufacturing. For example, embodiments of the fluid pump case can have a set of unique design changes, support structures in only coincidental locations, and an orientation definition to allow the pump case for the augmentor fuel pump to be additively manufactured without negatively impacting performance. Embodiments utilizing such a tilted orientation, design, and with only coincidental support structure and also enable reduced AM post processing cost, eliminate support removal activity required that is not incorporated in existing machine operations, allow for easy powder removal (e.g., no wells or small features), provide the same or less weight as traditional casted parts, and provide equal or better performance compared to traditional casted parts. Thus, pump cases can be additively manufactured while reducing cost and at least maintaining performance.

(57) In accordance with at least one aspect of this disclosure, referring additionally to FIGS. 2A-2E, an additively manufactured valve body **200** for a fluid pump case (e.g., pump case **100**) can include an interior surface **201** and a plurality of interior annular features **203a**, **203b**, **203c**, **203d**, **203e** integrally formed with and extending radially inwardly from the interior surface **201**. The interior surface **201** can form a substantially cylindrical shape, for example. The one or more annular features **203a**, **b**, **c**, **d**, **e** can be positioned apart in an axial direction (e.g., along centerline axis **205**) relative to each other, e.g., for forming a plurality of lands.

(58) The valve body **200** can include a support structure **207a**, **b**, **c**, **d** (e.g., a plurality thereof as shown) formed integrally with and connecting a lower-in-build annular feature (e.g., annular feature **203b**) to a higher-in-build annular feature (e.g., annular feature **203c**). For example, as shown in FIG. 2D, the relative higher and lower positions can be seen relative to the build plate **107**. The support structure **207a**, **b**, **c**, **d** can be any of the support structure **103a**, **b**, **c**, **d** as described above (e.g., support structure **103a** in FIG. 1M is the same as support structure **207d** shown in FIG. 2A).

(59) In certain embodiments, as best shown in FIG. 2A, the support structure **207** can include a filled-in Y-shape. Such a shape can allow for powder removal and cleaning, while providing adequate support and limiting waste material when removed, for example. The support structure **207a**, **b**, **c**, **d** can form openings **215a**, **215b**, for example, allowing powder removal. For example, the support structure **207a**, **b**, **c**, **d** can include a stem **209** and a wide portion **211** extending from the stem **209**. The stem **209** can be thinner than the wide portion **211**. The wide portion **211** can include a quasi-circular shape, or any other suitable shape, and can be solid (e.g., as shown) or have any suitable discontinuities.

(60) The stem **209** of the support structure **207a**, **b**, **c**, **d** can be integrally formed with and extend from a lower-in-build portion, e.g., portion **213a** of the lower-in-build annular feature (for example, feature **203c**). The wide portion **211** can extend from the stem **209** and can be integrally formed with and connected to a higher-in-build portion (e.g., portion **213b**) of the higher-in-build annular feature (e.g., feature **203d**).

(61) Any suitable number of support structures are contemplated herein (e.g., to provide support during build for any would-be overhanging features). The valve body **200** can include a support

structure **207a, b, c, d** connecting to each annular feature **203a, b, c, d, e** (e.g., one connecting each pair such that the total is one less than annular features). As shown, each support structure **207a, b, c, d** can extend between each successive pair of annular features **203a, b, c, d, e**. In this regard, the support structure **207a, b, c, d** can be milled away in the process of milling the annular features **203a, b, c, d, e** to their final shape and/or finish since all support structure can connect to the inner diameter of the annular features. An embodiment of a milled valve body **200** is shown in FIG. 2E.

(62) In certain embodiments, as shown in FIG. 2D, the valve body **200** can be constructed at an angle between the axial direction **205** and a surface of the build plate **107** of greater than about degrees. Any other suitable angle is contemplated herein (e.g., to allow suitable additive manufacturing of the valve body **200** (e.g., with the case **100** as described above). For example, the build angle can be such that no unsupported surface is less than about 20 degrees from horizontal.

(63) The valve body **200** can be additively manufacture as part of the case **100** as described above. Any other suitable application (e.g., in isolation of the case) is contemplated herein. In certain embodiments, the disclosed embodiments of support structure can be applied to any channel shaped body build at a suitable angle using additive manufacturing.

(64) In accordance with at least one aspect of this disclosure, an additively manufactured fluid pump case (e.g., case **100**) can include a valve body. The valve body can be any suitable valve body as disclosed herein, e.g., valve body **200** as described above. In accordance with at least one aspect of this disclosure, a method can include additively manufacturing a valve body (e.g., valve body **200**) of fluid pump case (e.g., fluid pump case **100**). The method can also include milling away the support structure **207a, b, c, d** while finish-milling the valve body **200**, for example.

(65) In certain embodiments, and as disclosed above with respect to FIGS. 1A-1P, additively manufacturing can be or include laser power bed fusion. Any other suitable process is contemplated herein (e.g., fused deposition, etc.). The method can include removing powder from within the valve body **200** before milling through openings **215a, 215b** defined by the support structure **207a, b, c, d**. The method can include inserting a valve (not specifically shown) into the valve body **200** (e.g., after being milled). For example, the valve can be a selector valve having a circumferentially defined flow path based on its axial and/or rotational position within the valve body **200**.

(66) This method can include any other suitable method(s) and/or portions thereof. Embodiments can include a means for additively manufacturing a valve body (e.g., valve body **200**).

(67) In certain additive manufacturing processes, there is a need for a powder flow path to remove powder from internal channels of parts. Embodiments disclosed herein provide support structure than can include a trunk extending upward and forward from a lower land, and a branch portion that contacts next land up from lower land. Certain fuel pumps are complex in shape and can benefit from embodiments of support structure disclosed in FIG. 2A-2E to provide additive manufacturing solutions that can save in powder removal costs by reducing powder removal time. Accordingly, embodiments of the support structure of FIGS. 2A-2E can provide a reduced additive manufacturing post processing cost, easy powder removal, increased volume air/powder/medium, reduced sintered feed stock material, no modification of machining process, and the same or better valve performance. Valve body geometry and orientation can be a performance consideration of the part, and the positioning thereof can be a function of the mating valve assembly's straight line action and associated parts.

(68) When valve bodies are at low build angles (e.g., caused by an overall orientation that is greater than about 45 degrees), they can have features that present build risk. Traditional redesigning solutions such as removing an overhanging surface and creating small powder removal holes drive up powder removal cost or are not feasible. By applying embodiments of the support structure disclosed (e.g., a "Y" support geometry) to connect surfaces that are to be machined, valve bodies that are to be built via additive manufacturing and oriented at about 45 degrees from horizontal can be supported while allowing ease of powder removal and minimizing sintered material that will become machined stock.

(69) Fuel pumps can use valves housed in such valve bodies as described above to regulate flow to primary and secondary systems of the pump. Valve bodies can have internal tubes that interface in the grooves (between annular features) of the body. The mating valve can interface with the lands of the valve body. Traditionally, these surfaces in the valve body have only be subtractively processed (e.g., machined/milled) to meet interface requirements regardless of manufacturing method. Any other suitable additive or subtractive post processing finishing method is contemplated herein.

(70) In accordance with at least one aspect of this disclosure, referring additionally to FIGS. 3A-3D, an additively manufactured valve body **300** for a fluid pump case (e.g., pump case **100**) can include an interior surface **301** defining a main channel **303** along an axis **305** (e.g., as shown in FIG. 3B) in an axial direction. The channel **303** can have a non-uniform cross-section configured to allow the valve body **301** to be additively manufactured without support structure, e.g., as shown in FIG. 3A.

(71) In certain embodiments, as can be seen in FIG. 1C, for example, the non-uniform cross-section can have a teardrop shape. For example, the interior surface **301** can include a curved lower-in-build portion **301a** and a ramped higher-in-build portion **301b** that forms a V-shape extending from the curved lower-in-build portion **301a**. As shown, the curved lower-in-build portion **301a** can have a semicircular cross-section. The ramped higher-in-build portion **301b** can include a curved peak **307** where the ramped surfaces meet. The ramped surfaces can form any suitable angle extending from the lower-in-build portion **301a** (e.g., about 30 to about 45 degrees). In certain embodiments, the valve body **300** can be additively manufactured such that the axis **305** forms an angle of about 45 degrees or less with a surface of the build plate **107** on which the valve body **300** is built.

(72) The interior surface **301** can further define one or more annular grooves **309** extending radially outwardly from the main channel **303**. The one or more annular grooves **309** can include an asymmetric groove profile. For example, a higher-in-build portion **309a** of the one or more annular grooves **309** can include a curved shape in the axial direction (e.g., U-shaped cross-section as shown in FIG. 3A), and a lower-in-build portion **309b** of the one or more annular grooves **309** can include a straight shape (e.g., rectilinear shaped cross-section as shown in FIG. 3A) in the axial direction.

(73) In certain embodiments, the one or more annular grooves **309** can be two annular grooves **309** positioned apart in an axial direction relative to each other, e.g., as shown. Any suitable number of annular grooves **309** are contemplated herein.

(74) The internal shape of the valve body **300** can enable additive manufacturing of the valve body **300**, e.g., as part of the fluid pump case **100** in the orientation disclosed in relation to FIGS. 1A-1P without utilizing support structure. For example, the valve body **300** can be positioned less than about 45 degrees in the tilted build orientation as described above. The shape of the interior surface **301** can be milled after additive manufacturing into a uniform cross-sectional shape (e.g., a cylindrical shape) as shown in FIG. 3D (e.g., with milled lands for a valve assembly (e.g., similar as disclosed above in FIGS. 2A-2E). While the valve body **300** is shown integrated with the fluid pump case **100**, any suitable application (e.g., in isolation of the pump case **100**) is contemplated herein.

(75) In accordance with at least one aspect of this disclosure, an additively manufactured fluid pump case (e.g., case **100**) can include a valve body **300** as disclosed herein, e.g., as described above. In accordance with at least one aspect of this disclosure, a method of manufacturing a fluid pump case (e.g., case **100**) can include additively manufacturing a valve body **300** as disclosed herein, e.g., as described above, and machining the main channel **303** to have a different cross-section to allow the valve body **300** to receive one or more valve components while leaving the one or more annular grooves **309**.

(76) In accordance with at least one aspect of this disclosure, embodiments can include fluid pump case means for use as a fluid pump housing, the fluid pump case means having valve body means

configured to allow additive manufacturing of the fluid pump case without building support structure in the valve body.

(77) Embodiments can have tops of grooves with a round cross-section to allow a smooth build without low overhang while the bottoms of the grooves can have a square cross-section. The total volume of the grooves can be about the same as a traditional design to allow the performance to be the same or similar to traditional embodiments. Embodiments of the valve body **300** can have a teardrop shape that allows open area for power removal, for example. As presented above, embodiments can allow for building certain valve bodies (e.g., valve body **300** of a fluid pump case **100**) at build angles of less than about 45 degrees relative to the axis of channel **303**.

(78) Embodiments can include a teardrop self-supporting profile that also provide ample powder removal flow area (e.g., the whole of teardrop shaped channel **303** after additive manufacturing and before milling). By applying a unique teardrop profile to the valve body **300**, coupled with an asymmetric groove using a semi-circular cross section swept into a rectangular cross section, the low angle-in-build valve body can be self-supporting without restricting powder flow.

(79) In accordance with at least one aspect of this disclosure, referring additionally to FIGS. **4A-4G**, a fluid pump case, e.g., fluid pump case **100**, can include an interior volute surface **401** defining a volute channel **400**. The interior volute surface **401** can include a first portion **401a** having a D-shaped cross-section and a second portion **401b** having a teardrop shaped cross-section downstream of the first portion **401a**. The interior volute surface can include a transition portion **401c** between the first portion **401a** and the second portion **401b** having a changing cross-section (e.g., that morphs from a D-shaped cross-section to a teardrop shaped cross-section while curving as shown). The transition portion **401c** can be defined between lines **409a** and **409b**, e.g., as shown in FIGS. **4A** and **4D**.

(80) The teardrop cross-section can include a curved lower-in-build portion **403a** and a ramped higher-in-build portion **403b** that forms a V-shape extending from the curved lower-in-build portion **403a**. In certain embodiments, the teardrop cross-sectional shape can be the same or similar to that disclosed above with respect to the valve body **300** of FIG. **3A**. For example, the curved lower-in-build portion **401a** can have a semicircle cross-section, e.g., as shown. The ramped higher-in-build portion can have a curved peak **405**. In certain embodiments, the teardrop cross-section can be aligned with the teardrop shape of the valve body **300**.

(81) The second portion **401b** can be shaped such that it is additively manufactured without support structure, e.g., as shown. For example, a centerline of the peak **405** (e.g., as shown in FIGS. **4A** and **4C**) along the length of the second portion **401b** can be coplanar with a vertical build direction (and a centerline axis of the volute channel **400**) in build such that the teardrop shape is symmetric about a vertical plane (e.g., on which the vertical build axis lays), e.g., as illustrated in FIG. **4G**. Any suitable variation or asymmetry that still allows building without support structure is contemplated herein. As shown, the teardrop shape is rotated about 45 (e.g., about 43.6) degrees relative to a pump centerline axis.

(82) As shown, the first portion **401a** can have an expanding flow area in a flow direction toward the second portion **401b**. The second portion **401b** can have an expanding flow area or constant flow area in the flow direction, for example. In certain embodiments, the volute channel **400** can consistently expand from the first portion **401a** through the second portion **401b**.

(83) The volute channel **400** can further include a reducing portion **407** downstream of the second portion **401b**. The reducing portion **407** can include a teardrop shape cross-section also, or any other suitable shape to allow building without support structure. The reducing portion **407** can be connected to a valve channel (e.g., channel **303**) defined by a valve body (e.g., valve body **300**). Embodiments can be utilized with the above disclosed fluid pump case **100** or any other suitable application (e.g., in isolation from fluid pump **100**).

(84) Embodiments can be oriented such that the peak **405** of the teardrop is positioned relative to the build area such that neither ramped wall forming the ramped higher-in-build portion will be less

than about 20 degrees from horizontal, for example. The peak **405** of the teardrop shape can be in any suitable orientation as long as the build orientation of the volute channel **400** is changed to have the ramped portion where an overhang would be. The shown position of the teardrop shape is a suitable position for the tilted orientation described above with respect to FIGS. **1A-1P**. The above embodiments of valve bodies **200**, **300** are also shown shaped and oriented (with or without build structure as described above) to allow for building in the tilted orientation described above with respect to FIGS. **1A-1P**.

(85) Unlike certain embodiments of the valve bodies **200**, **300** disclosed above, the interior surface **401** need not be milled to a final shape and can maintain the teardrop shape in the final device. While the teardrop structure allows building without support structure, it also maintains strength under pressure. Such a shape is acceptable in use without compromising function. Embodiments can include transitions before and after the teardrop that are smooth to aid in powder removal, for example, as opposed to traditional embodiments that have sharp angles that are machined away. This also reduces post processing required since the reducing portion of the volute channel is already formed to the desired shape.

(86) In accordance with at least one aspect of this disclosure, a method can include additively manufacturing a fluid pump case to include an interior volute surface **401** defining a volute channel **400** as disclosed herein, e.g., as described above. Additively manufacturing can include aligning a centerline of the peak **405** along the length of the second portion **401b** to be coplanar with a vertical build direction in build such that the teardrop shape is symmetric about a vertical plane. In accordance with at least one aspect of this disclosure, embodiments can include volute means for a fluid pump case.

(87) Fuel pumps use volutes designed for the casting manufacturing process with input from CFD models, for example. The part can be surface finish critical, with limited to no line of sight to feature geometry. Volute design can be critical to the function of a pump as it collects and slows the flow while maintaining minimal pressure from the inlet impeller assembly. To maintain part performance, traditional volute geometry cannot be used if the pump case is to be additively manufacture while also maintaining a cost practicality.

(88) The fuel pump volute is composed of a fluid passage that receives fuel from a series passages connected to the impeller volume, as appreciated by those having ordinary skill in the art and as shown in the above described figures (e.g., all connected to the first portion **401a** and/or transition portion **401c**). In pump embodiments, e.g., as shown in FIG. **1P**, the volute transmits the fluid from the inlet impeller assembly to the valve system (e.g., associated with valve body **300**) that connects to a fluid (e.g., fuel) discharge while maintaining minimal loss of pressure.

(89) The volute is highly sensitive to pressure drop, which is directly affected by cross-sectional area, cross-sectional shape, volume, path, volume expansion rate, and surface roughness. Typical devices can include a semi rectangular “D” cross-section that transitions into a circle shape, however, this cannot be properly additively manufactured (e.g., at least in the tilted orientation disclosed above with respect to FIGS. **1A-1P**) at least due to the fact that support structure would be required and would be difficult if not impossible to remove.

(90) Embodiments include a teardrop non-uniform pump volute that is a self-supporting volute via a unique volute profile and shape. Embodiments can include a volute profile can transition from a CFD driven “D” shaped cross section to a teardrop shape based on part orientation. This customized feature shape driven by the part orientation can allow the fuel pump to be additively manufactured with no need for support structure, and therefore no support removal costs. Embodiments also minimize high surface roughness areas having the teardrop oriented with the build direction. Embodiments also provide a smoother volute to valve transition and powder removal and flow benefit.

(91) Embodiments provide reduced additive manufacturing post processing costs, no support structure in the volute, easy powder removal (e.g., no wells or small features), weight neutral

lighter pump cases, a more efficient structural transition from the D shaped cross-section to the teardrop shape compared to traditional D-to-round transitions, equal or better performance, no integral supports or additional material obstructing flow paths, approximately equivalent volute cross-sectional area (e.g., within 3%), approximately equivalent volute volume equivalent (e.g., within 3%), minimal area of surface roughness (e.g., less than 400, arched downskin), and a smoother volute to valve flowpath.

(92) While certain figures may include embodiments of dimensions and relative dimensions, such dimensions and relative dimensions are not limiting. Any suitable dimensions or relative dimensions to allow additive manufacturing and/or for any suitable application are contemplated herein. Any suitable size or scale of the embodiments shown is contemplated herein.

(93) Those having ordinary skill in the art understand that any numerical values disclosed herein can be exact values or can be values within a range. Further, any terms of approximation (e.g., “about”, “approximately”, “around”) used in this disclosure can mean the stated value within a range. For example, in certain embodiments, the range can be within (plus or minus) 20%, or within 10%, or within 5%, or within 2%, or within any other suitable percentage or number as appreciated by those having ordinary skill in the art (e.g., for known tolerance limits or error ranges).

(94) The articles “a”, “an”, and “the” as used herein and in the appended claims are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article unless the context clearly indicates otherwise. By way of example, “an element” means one element or more than one element.

(95) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

(96) As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.”

(97) Any suitable combination(s) of any disclosed embodiments and/or any suitable portion(s) thereof are contemplated herein as appreciated by those having ordinary skill in the art in view of this disclosure.

(98) The embodiments of the present disclosure, as described above and shown in the drawings, provide for improvement in the art to which they pertain. While the subject disclosure includes reference to certain embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the spirit and scope of the subject disclosure.

Claims

1. A fluid pump case, comprising: an interior volute surface defining a volute channel, comprising: a first portion having a D-shaped cross-section defined entirely by the case; a second portion having a teardrop shaped cross-section downstream of the first portion; and a transition portion between the first portion and the second portion having a changing cross-section, wherein the second portion is shaped such that it is additively manufactured without support structure.
2. The fluid pump case of claim 1, wherein the teardrop cross-section includes a curved lower-in-build portion and a ramped higher-in-build portion that forms a V-shape extending from the curved lower-in-build portion.
3. The fluid pump case of claim 2, wherein the curved lower-in-build portion has a semicircle cross-section.
4. The fluid pump case of claim 3, wherein the ramped higher-in-build portion includes a curved peak.
5. The fluid pump case of claim 4, wherein a centerline of the peak along the length of the second portion is coplanar with a vertical build direction in build such that the teardrop shape is symmetric about a vertical plane.
6. The fluid pump of claim 1, wherein the first portion has an expanding flow area in a flow direction toward the second portion.
7. The fluid pump of claim 6, wherein the second portion has an expanding flow area or constant flow area in the flow direction.
8. The fluid pump of claim 7, wherein the volute channel further includes a reducing portion downstream of the second portion.
9. The fluid pump of claim 8, wherein the reducing portion includes a teardrop shape cross-section.
10. The fluid pump of claim 9, wherein the reducing portion is connected to a valve channel defined by a valve body.
11. The fluid pump case of claim 10, wherein the valve body comprises: an interior surface defining a main channel along an axis in an axial direction, wherein the main channel has a non-uniform cross-section, wherein the non-uniform cross-section has a teardrop shape, wherein the teardrop shape cross section of the main channel of the valve body is aligned with the teardrop shape cross section of the interior volute surface of the fluid pump case.
12. The fluid pump case of claim 11, wherein the interior surface of the valve body further defines one or more annular grooves extending radially outwardly from the main channel, wherein the one or more annular grooves include an asymmetric groove profile.
13. A method, comprising: additively manufacturing a fluid pump case to include an interior volute surface defining a volute channel, comprising: a first portion having a D-shaped cross-section defined entirely by the case; a second portion having a teardrop shaped cross-section downstream of the first portion; and a transition portion between the first portion and the second portion having a changing cross-section, wherein the second portion is shaped such that it is additively manufactured without support structure.
14. The method of claim 13, wherein the teardrop cross-section includes a curved lower-in-build portion and a ramped higher-in-build portion that forms a V-shape extending from the curved lower-in-build portion.
15. The method of claim 14, wherein the curved lower-in-build portion has a semicircle cross-section.
16. The method of claim 15, wherein; the ramped higher-in-build portion includes a curved peak; and the additively manufacturing includes aligning a centerline of the peak along the length of the second portion to be coplanar with a vertical build direction in build such that the teardrop shape is symmetric about a vertical plane.

17. A fluid pump, comprising: a fluid pump case of claim 1; and volute means for a flowing volute flow through the fluid pump case.
18. A fluid pump case, comprising: an interior volute surface defining a volute channel, comprising: a first portion having a D-shaped cross-section defined entirely by the case; a second portion having a teardrop shaped cross-section downstream of the first portion; and a transition portion between the first portion and the second portion having a changing cross-section, wherein the second portion is shaped to permit it to be additively manufactured without support structure.
19. The fluid pump case of claim 18 wherein: in a first orientation of the pump case, the D-shape is oriented with its stem inward and the teardrop is oriented with its vertex inward and upward.
20. The fluid pump case of claim 1, wherein: the volute channel further includes a reducing portion downstream of the second portion; the reducing portion is connected to a valve channel defined by a valve body; and the valve body comprises: an interior surface defining a main channel along an axis in an axial direction, wherein the main channel has a non-uniform cross-section, wherein the non-uniform cross-section has a teardrop shape, wherein the teardrop shape cross section of the main channel of the valve body is aligned with the teardrop shape cross section of the interior volute surface of the fluid pump case.
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